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# MLOps Group Assignment

AG News Text Classification Pipeline

Fine-tuning DistilBERT with Docker, CI/CD, and W&B Tracking

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**Course:** MLOps — PGD AI Program  
**Institute:** IIT Jodhpur  
**Group Number:** Group 13  
**Submission Date:** June 2026

| Name                       | Roll Number | Contribution                            |
|----------------------------|-------------|-----------------------------------------|
| Ritesh Maury               | G25AIT2087  | GitHub Setup, Project Lead, Merges      |
| Kosuru Yuvaraj             | G25AIT2054  | Training, Docker, CI/CD, HF Hub, Report |
| Laishram Khumanleima Chanu | G25AIT2056  | Data Preparation, Inference Script      |
| Guntakal Sandeep Kumar     | G25AIT2036  | W&B Tracking, Evaluation, Testing       |

## All Submission Links

**GitHub Repository:** [riteshmaury-iitj/group13-assignment-mlops](https://github.com/riteshmaury-iitj/group13-assignment-mlops)  
**Kaggle Notebook V1:** [kyuvarajg25ait2054/group13-ag-news-k](https://www.kaggle.com/kyuvarajg25ait2054/group13-ag-news-k)  
**Kaggle Notebook V2:** [kyuvarajg25ait2054/group13-ag-news-k](https://www.kaggle.com/kyuvarajg25ait2054/group13-ag-news-k)  
**HuggingFace Model:** [YuvarajK-g25ait2054/ag-news-distilbert](https://huggingface.co/YuvarajK-g25ait2054/ag-news-distilbert)  
**Docker Image:** [yuvarajkg25ait2054/ag-news-classifier](https://hub.docker.com/r/yuvarajkg25ait2054/ag-news-classifier)  
**W&B Dashboard:** [g25ait2054-iit-jodhpur/mlops-assignment3](https://wandb.ai/g25ait2054-iit-jodhpur/mlops-assignment3)

## Member Profiles & Project Links

All links are public and live at submission time.

### Member Profiles

| Member                             | GitHub                           | Kaggle                             | HuggingFace                         | W&B                                |
|------------------------------------|----------------------------------|------------------------------------|-------------------------------------|------------------------------------|
| Ritesh Maury<br>G25AIT2087         | <a href="#">riteshmaury-iitj</a> | <a href="#">ritesh329</a>          | <a href="#">Recurrent</a>           | <a href="#">models-iit-jodhpur</a> |
| Kosuru Yuvaraj<br>G25AIT2054       | <a href="#">yuvarajkosuru</a>    | <a href="#">kyuvarajg25ait2054</a> | <a href="#">YuvarajK-g25ait2054</a> | <a href="#">g25ait2054</a>         |
| Laishram K.<br>Chanu<br>G25AIT2056 | <a href="#">group13-mlops</a>    | <a href="#">khumanleima</a>        | <a href="#">LKHUMANLEIMA</a>        | <a href="#">g25ait2056</a>         |
| Guntakal Sandeep<br>G25AIT2036     | <a href="#">Sandeep_dev</a>      | <a href="#">guntakalsandeep</a>    | <a href="#">guntaksa</a>            | <a href="#">g25ait2036</a>         |

### Project Links (all public)

| Component                            | Link                                                                   |
|--------------------------------------|------------------------------------------------------------------------|
| GitHub Repository (public)           | <a href="#">github.com/riteshmaury-iitj/group13-assignment-mlops</a>   |
| Kaggle Notebook — Version 1 (public) | <a href="#">kaggle.com/code/kyuvarajg25ait2054/group13-ag-news-k</a>   |
| Kaggle Notebook — Version 2 (public) | <a href="#">kaggle.com/code/kyuvarajg25ait2054/group13-ag-news-k</a>   |
| HuggingFace Model (public)           | <a href="#">huggingface.co/YuvarajK-g25ait2054/ag-news-distilbert</a>  |
| Docker Image (publicly pullable)     | <a href="#">hub.docker.com/r/yuvarajkg25ait2054/ag-news-classifier</a> |
| W&B Project Dashboard (public)       | <a href="#">wandb.ai/g25ait2054-iit-jodhpur/mlops-assignment3</a>      |

## 1. GitHub Repository, Data Preparation and Model Selection (Tasks 1–3)

### 1.1 Task 1 — GitHub Repository Setup [10 marks]

Repository [riteshmaurya-iitj/group13-assignment-mlops](https://github.com/riteshmaurya-iitj/group13-assignment-mlops) is public with three branches: **main** (protected, PRs only), **develop** (integration), and **feature/\*** (short-lived). Branch protection on **main** requires at least one approval before merging. All work was merged via Pull Request from **develop**.

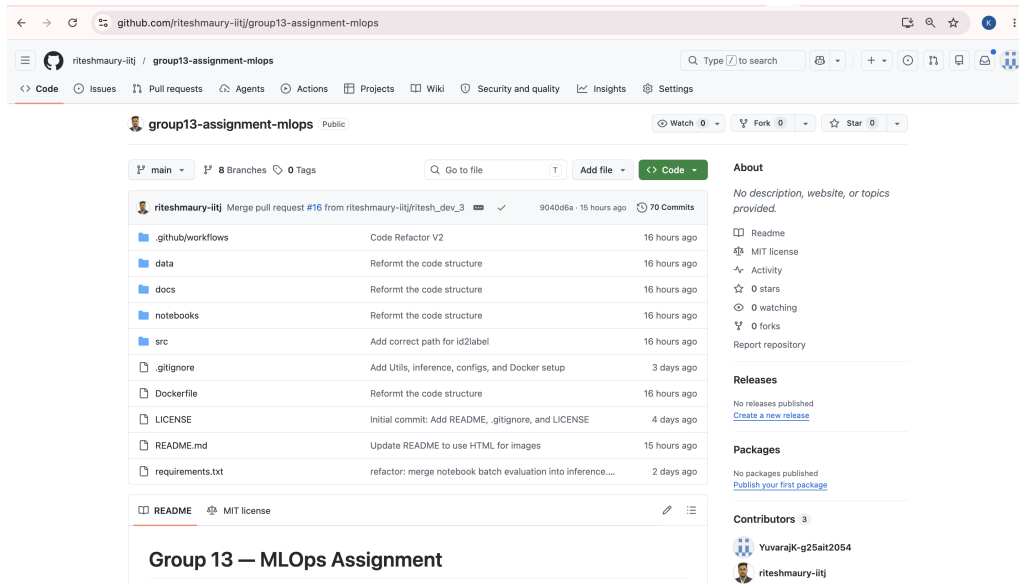


Figure 1: GitHub repository — public visibility and branch list

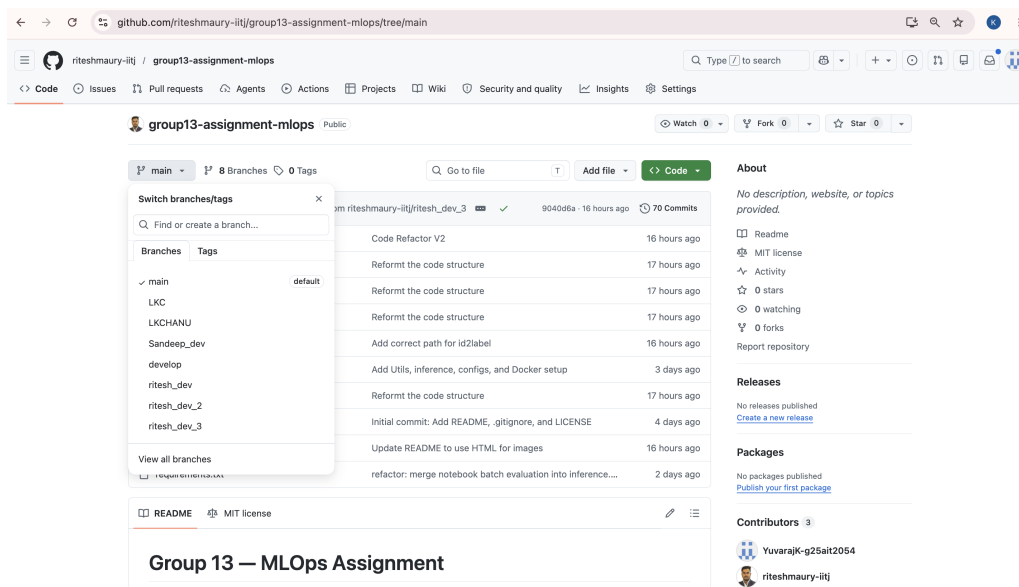


Figure 2: GitHub Settings — Branch protection rule and repo details

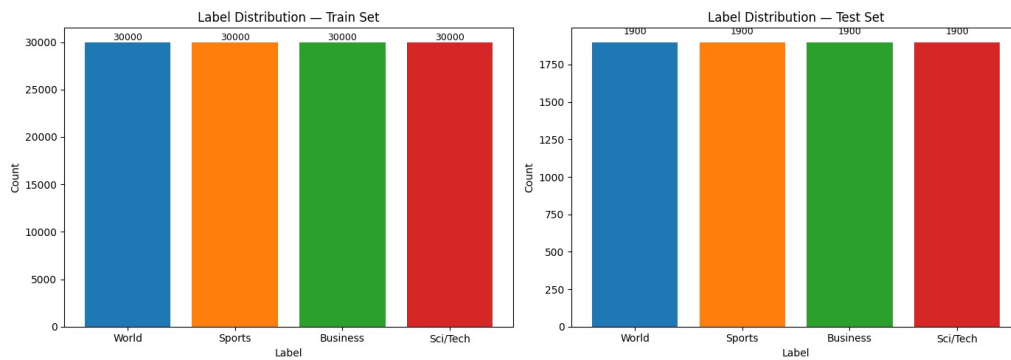


Figure 3: GitHub Settings — Collaborators: g25ait2036, g25ait2056-beep, YuvarajK-g25ait2054

## 1.2 Task 2 — Data Preparation & Normalisation [15 marks]

Dataset: `ag_news` from HuggingFace (120,000 train / 7,600 test, 4 balanced classes). Preprocessing applied in `prepare_data.ipynb`:

- Lowercase, remove URLs, strip HTML tags, remove punctuation, normalize whitespace.
- Label mapping saved to `id2label.json`: {0: World, 1: Sports, 2: Business, 3: Sci/Tech}.
- Tokenized with `max_length=128`.

screenshots/label\_dist.png

Figure 4: Test set label distribution — 4 balanced classes, 1,900 samples each (7,600 total)



Figure 5: Test set text length distribution — word count per class (mean: 64 words, max: 200)

### 1.3 Task 3 — Model Selection [10 marks]

Selected **distilbert-base-uncased**: 66M parameters, retains ~97% of BERT performance, and trains on AG News in ~40 min on a T4 GPU. Loaded via `AutoModelForSequenceClassification` with `num_labels=4` and `id2label/label2id` mappings. Tokenizer uses `truncation=True`, `padding=True`, `max_length=128`.

## 2. Training, W&B Tracking and HuggingFace Deployment (Tasks 4–5)

### 2.1 Task 4 — Training on Kaggle with W&B [25 marks]

Training on Kaggle (GPU T4 ×2), tracked via W&B project `mlops-assignment3`. Secrets (W&B, HuggingFace, GitHub) stored as Kaggle Secrets, loaded via `UserSecretsClient`.

#### Hyperparameter Comparison:

| Hyperparameter      | Version 1 (run-v1)      | Version 2 (run-v2)      |
|---------------------|-------------------------|-------------------------|
| Base Model          | distilbert-base-uncased | distilbert-base-uncased |
| Learning Rate       | 2e-5                    | 5e-5                    |
| Batch Size (train)  | 16                      | 64                      |
| Epochs              | 3                       | 5                       |
| Weight Decay        | 0.01                    | 0.01                    |
| Max Sequence Length | 128                     | 128                     |
| Evaluation Strategy | Per epoch               | Per epoch               |
| Best Model Metric   | F1 (weighted)           | F1 (weighted)           |

### Results:

| Version            | Accuracy      | F1 (weighted) | Eval Loss |
|--------------------|---------------|---------------|-----------|
| Version 1 (run-v1) | <b>94.34%</b> | <b>0.9434</b> | 0.3679    |
| Version 2 (run-v2) | 94.12%        | 0.9412        | 0.3575    |

Version 1 achieved higher accuracy and F1; selected as the final model. Both versions exceeded 94% accuracy.

← Viewing Version 5: 📌 Save & Run All • June 8, 2026 at 2:19 PM

```
In [10]: # Evaluate v1
results_v1 = trainer_v1.evaluate()
print(f"\n=== Version 1 Results ===")
print(f" Accuracy: {results_v1['eval_accuracy']:.4f}")
print(f" F1 Score: {results_v1['eval_f1']:.4f}")
print(f" Eval Loss: {results_v1['eval_loss']:.4f}")

# Finish W&B run
wandb.finish()
print("\nW&B run-ver1 finished.")

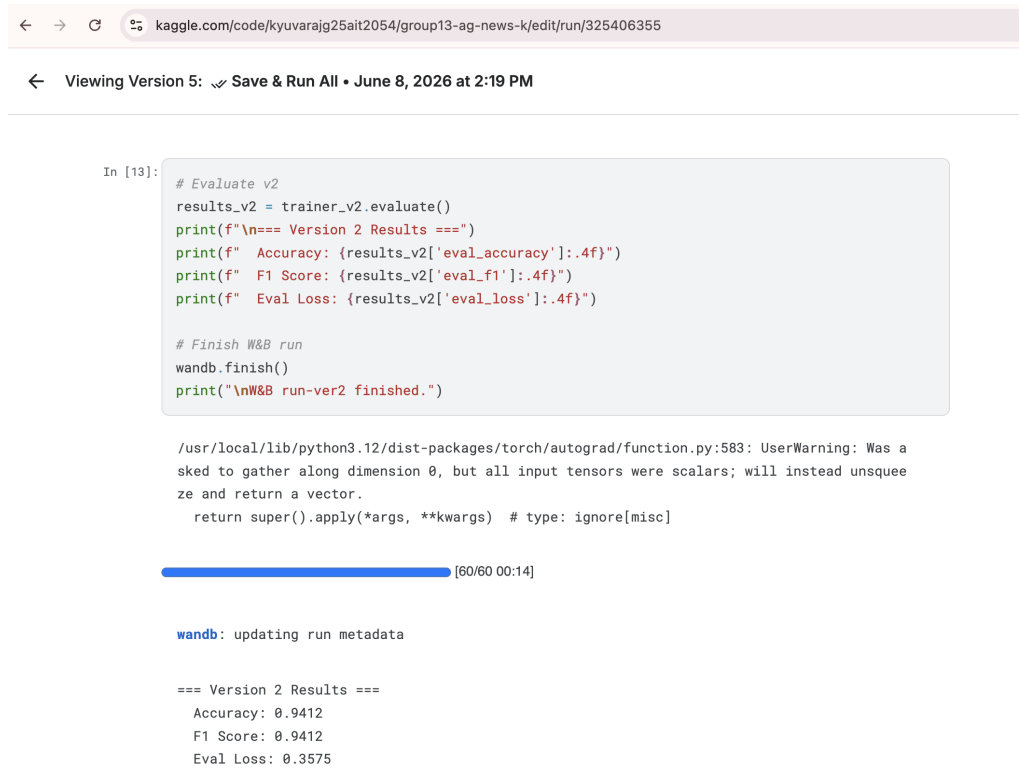
/usr/local/lib/python3.12/dist-packages/torch/autograd/function.py:583: UserWarning: Was a
sked to gather along dimension 0, but all input tensors were scalars; will instead unsquee
ze and return a vector.
  return super().apply(*args, **kwargs) # type: ignore[misc]

[119/119 00:14]

wandb: updating run metadata

=== Version 1 Results ===
Accuracy: 0.9434
F1 Score: 0.9434
Eval Loss: 0.3679
```

Figure 6: Kaggle notebook — Version 1 training output (eval accuracy 94.34%, F1 0.9434)



```

In [13]: # Evaluate v2
results_v2 = trainer_v2.evaluate()
print(f"\n=== Version 2 Results ===")
print(f" Accuracy: {results_v2['eval_accuracy']:.4f}")
print(f" F1 Score: {results_v2['eval_f1']:.4f}")
print(f" Eval Loss: {results_v2['eval_loss']:.4f}")

# Finish W&B run
wandb.finish()
print("\nW&B run-ver2 finished.")

/usr/local/lib/python3.12/dist-packages/torch/autograd/function.py:583: UserWarning: Was a
sked to gather along dimension 0, but all input tensors were scalars; will instead unsquee
ze and return a vector.
  return super().apply(*args, **kwargs) # type: ignore[misc]

[60/60 00:14]

wandb: updating run metadata

=== Version 2 Results ===
Accuracy: 0.9412
F1 Score: 0.9412
Eval Loss: 0.3575

```

Figure 7: Kaggle notebook — Version 2 training output (eval accuracy 94.12%, F1 0.9412)

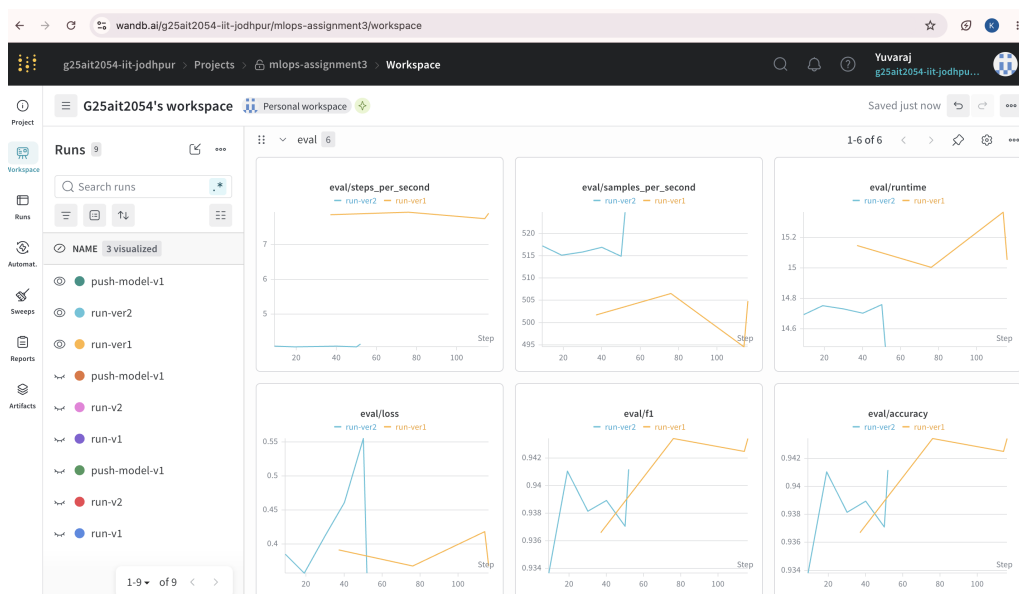
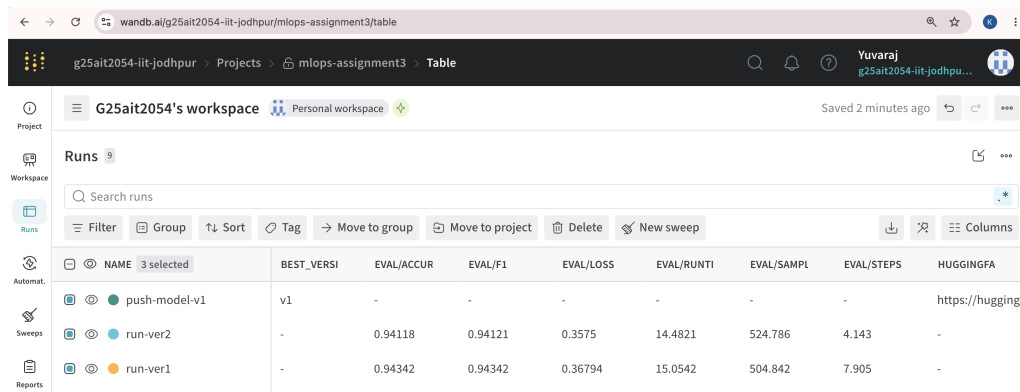


Figure 8: W&amp;B dashboard — mlops-assignment3 showing run-v1 and run-v2 with metrics



| NAME          | BEST_VERSION | EVAL/ACCUR | EVAL/F1 | EVAL/LOSS | EVAL/RUNTIME | EVAL/SAMPLE | EVAL/STEPS | HUGGINGFA        |
|---------------|--------------|------------|---------|-----------|--------------|-------------|------------|------------------|
| push-model-v1 | v1           | -          | -       | -         | -            | -           | -          | https://huggingf |
| run-ver2      | -            | 0.94118    | 0.94121 | 0.3575    | 14.4821      | 524.786     | 4.143      | -                |
| run-ver1      | -            | 0.94342    | 0.94342 | 0.36794   | 15.0542      | 504.842     | 7.905      | -                |

Figure 9: W&amp;B comparison chart — eval/accuracy and eval/loss across both runs

## 2.2 Task 5 — Push Model to HuggingFace Hub [5 marks]

Version 1 pushed to HuggingFace Hub via `trainer.push_to_hub()`. Model is public at:

<https://huggingface.co/YuvarajK-g25ait2054/ag-news-distilbert>

Model card (MODEL\_CARD.md) covers description, hyperparameters, evaluation results, and usage.

huggingface.com/YuvarajK-g25ait2054/ag-news-distilbert/tree/main

Hugging Face

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YuvarajK-g25ait2054

ag-news-distilbert

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PyTorch

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English

distilbert

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news-classification

Eval Results (legacy)

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Model card

Files and versions xet

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269 MB

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1 contributor

History: 11 commits

Contribute

YuvarajK-g25ait2054

Update README.md

76a7912

VERIFIED

1 day ago

.gitattributes

Safe

1.52 kB

initial commit

3 days ago

README.md

7.37 kB

Update README.md

1 day ago

config.json

Safe

853 Bytes

Upload DistilBertForSequenceClassificati...

3 days ago

model.safetensors

Safe

268 MB

xet

Upload DistilBertForSequenceClassificati...

3 days ago

tokenizer.json

Safe

712 kB

Upload tokenizer

3 days ago

tokenizer\_config.json

Safe

322 Bytes

Upload tokenizer

3 days ago

Figure 10: HuggingFace model page — YuvarajK-g25ait2054/ag-news-distilbert

## 3. Docker Deployment and GitHub Actions CI/CD (Tasks 6–7)

### 3.1 Task 6 — Dockerfile [10 marks]

Docker image hosted at [yuvarajkg25ait2054/ag-news-classifier:latest](https://hub.docker.com/r/yuvarajkg25ait2054/ag-news-classifier:latest).

- Base: `python:3.10-slim`; PyTorch 2.5.1+cpu from PyTorch's CPU wheel index.
- Model pre-baked into image during `docker build` — no internet required at runtime.
- Default CMD runs `--demo` mode: classifies all 4 sample texts from `inference.ipynb` Cell 7.

# Demo mode (default) -- runs all 4 notebook texts:

```
docker run --rm yuvarajkg25ait2054/ag-news-classifier:latest
```

# Single text:

```
docker run --rm yuvarajkg25ait2054/ag-news-classifier:latest \
```



```
--text "Stock market rises on strong earnings"
```

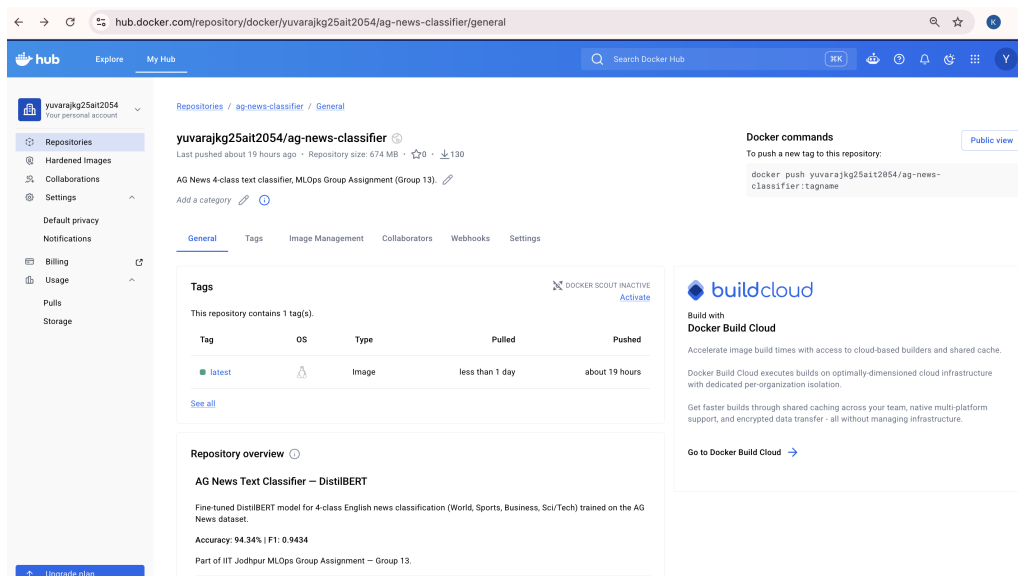


Figure 11: Docker Hub — yuvarajkg25ait2054/ag-news-classifier (publicly pullable)

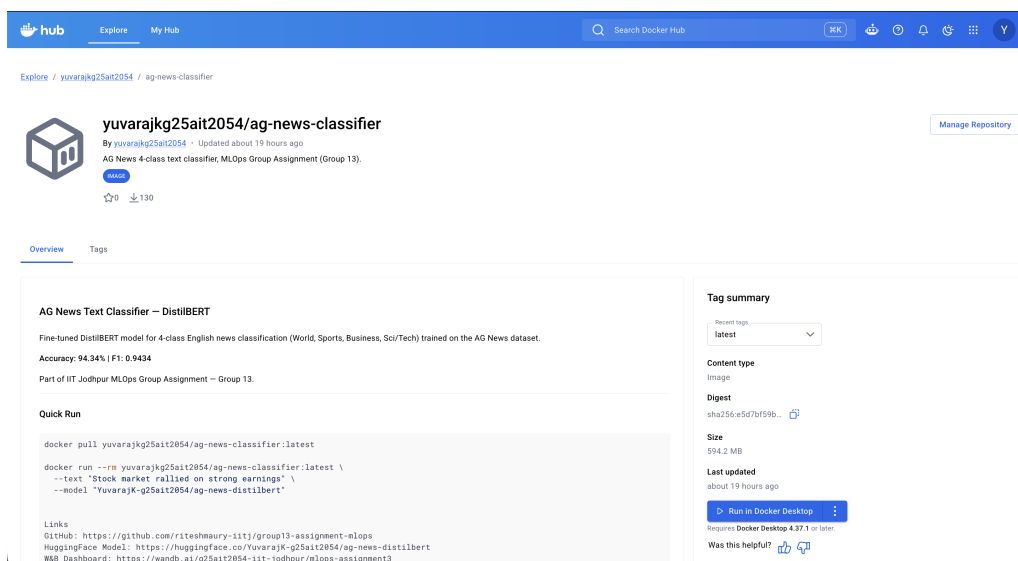


Figure 12: Docker container — terminal output showing inference results for all 4 categories

### 3.2 Task 7 — GitHub Actions Workflows [15 marks]

Two workflows under `.github/workflows/`:

- **ci.yml** — triggered on push/PR to `develop` and `main`. Validates required files, runs Python syntax check (`py_compile`), and validates JSON files.
- **inference.yml** — installs dependencies and runs `inference.py` end-to-end with a hard-coded test sentence against the HuggingFace model.

Both workflows pass with green checkmarks.

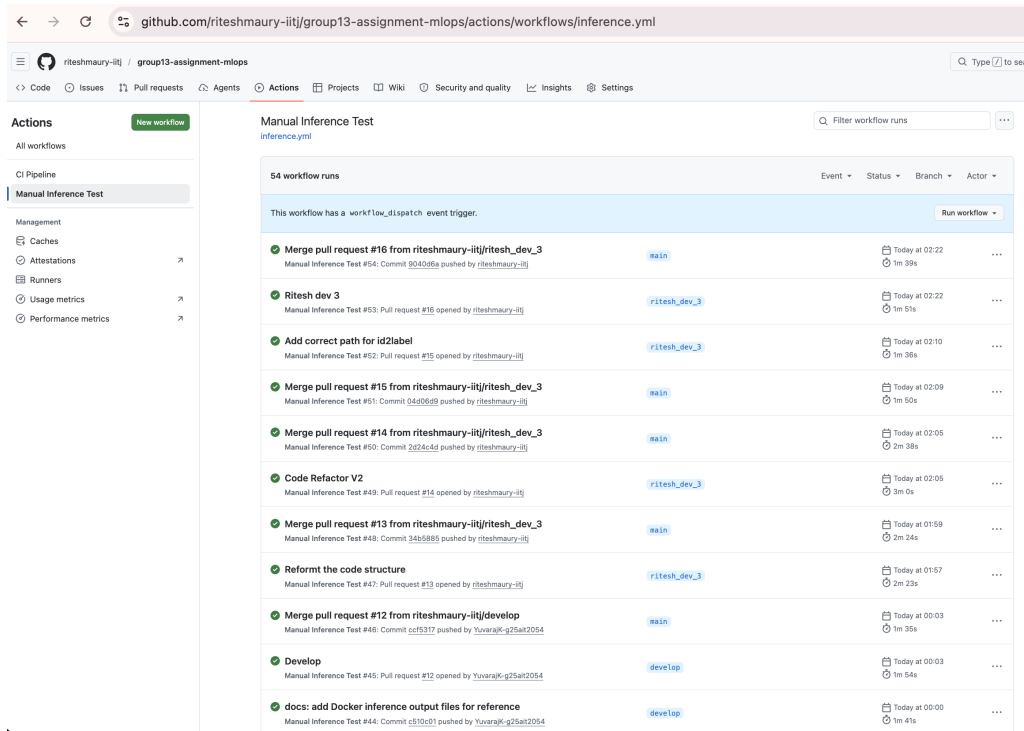


Figure 13: GitHub Actions — CI Pipeline and Inference workflow with green checkmarks

## 4. W&B Experiment Comparison and Conclusion (Task 8)

### 4.1 Task 8 — W&B Experiment Comparison [5 marks]

The W&B project is publicly accessible at:

<https://wandb.ai/g25ait2054-iiit-jodhpur/mlops-assignment3>

The project contains runs `run-v1`, `run-v2`, and `push-model-v1` (logs HuggingFace model URL to W&B summary).

- Version 1 ( $lr=2e-5$ ,  $bs=16$ ) converged smoothly; Version 2 ( $lr=5e-5$ ,  $bs=64$ ) had lower eval loss but slightly worse F1.
- Accuracy difference is small (0.22%); Version 1 selected for deployment based on higher F1.

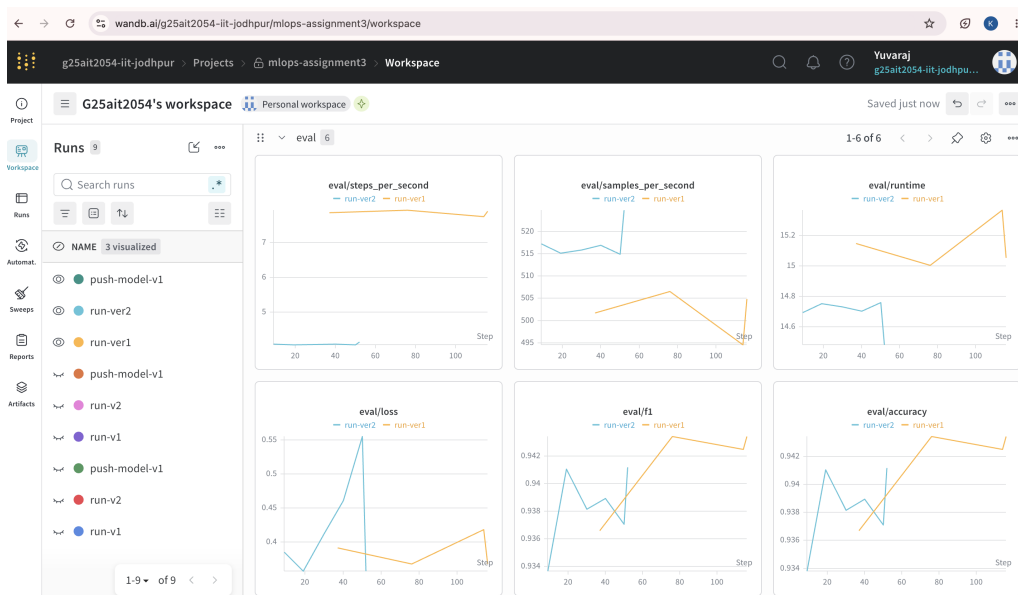


Figure 14: W&amp;B runs table — run-v1 and run-v2 with accuracy, F1, and loss

|               | BEST_VERSION | EVAL/ACCUR | EVAL/F1 | EVAL/LOSS | EVAL/RUNTIME | EVAL/SAMPLE | EVAL/STEPS | HUGGINGFACE                                                                                                                   |
|---------------|--------------|------------|---------|-----------|--------------|-------------|------------|-------------------------------------------------------------------------------------------------------------------------------|
| push-model-v1 | v1           | -          | -       | -         | -            | -           | -          | <a href="https://huggingface.co/g25ait2054-ilt-jodhpur/ml-models">https://huggingface.co/g25ait2054-ilt-jodhpur/ml-models</a> |
| run-ver2      | -            | 0.94118    | 0.94121 | 0.3575    | 14.4821      | 524.786     | 4.143      | -                                                                                                                             |
| run-ver1      | -            | 0.94342    | 0.94342 | 0.36794   | 15.0542      | 504.842     | 7.905      | -                                                                                                                             |

Figure 15: W&amp;B comparison view — eval/accuracy and eval/loss with both runs overlaid

## 4.2 Challenges

- **Docker PyTorch:** Default PyPI torch==2.5.1 has no CPU wheel; fixed by using PyTorch's CPU index URL (2.5.1+cpu).
- **DistilBERT token\_type\_ids:** DistilBERT does not use segment embeddings; had to remove token\_type\_ids from tokenizer output before model call.
- **GitHub Actions inputs:** github.event.inputs is empty on push triggers; fixed by hardcoding test values in the workflow YAML.
- **Merge conflicts:** Resolved by keeping develop versions of conflicted files.

### 4.3 Summary of Deliverables

| Task   | Description                 | Status | Link / Artefact                      |
|--------|-----------------------------|--------|--------------------------------------|
| Task 1 | GitHub Repo + Branching     | Done   | <a href="#">GitHub</a>               |
| Task 2 | Data Prep + Normalisation   | Done   | <code>prepare_data.ipynb</code>      |
| Task 3 | Model Selection             | Done   | <code>distilbert-base-uncased</code> |
| Task 4 | Training (v1 + v2) + W&B    | Done   | <a href="#">Kaggle</a>               |
| Task 5 | HuggingFace Hub Push        | Done   | <a href="#">HF Hub</a>               |
| Task 6 | Docker Image                | Done   | <a href="#">Docker Hub</a>           |
| Task 7 | GitHub Actions (CI + Infer) | Done   | <a href="#">Actions</a>              |
| Task 8 | W&B Comparison              | Done   | <a href="#">W&amp;B</a>              |